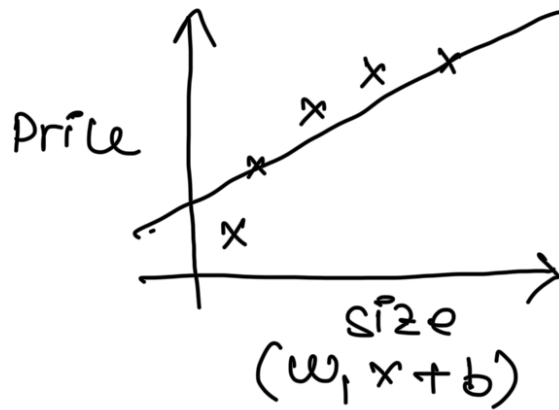


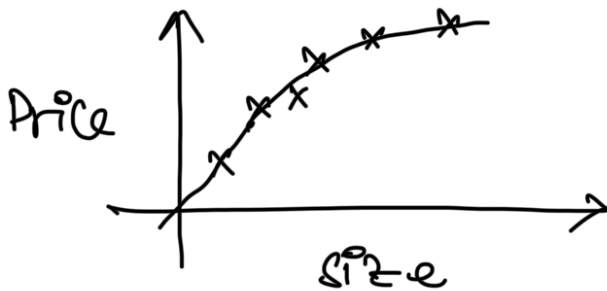
Linear Regression



- does not fit training data well

⇒ Underfitting
or
High bias.

⇓ Too few Parameters to Predict.



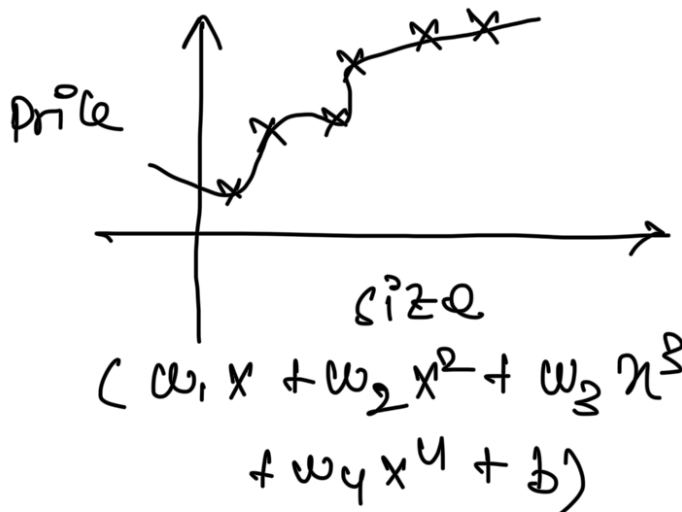
$$(w_1 x + w_2 x^2 + b)$$



Appropriate amount of parameters.

- fits training set pretty well

⇒ Generalization



Too many parameters

- fits the training set extremely well

- But not able to predict output well.

• Overfitting or
High variance.

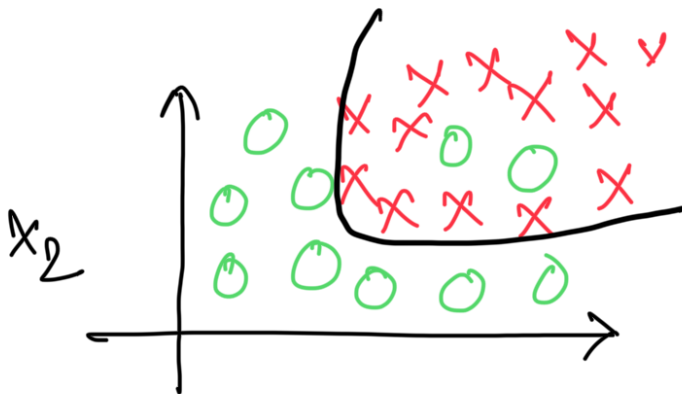
Parameters.

Classification



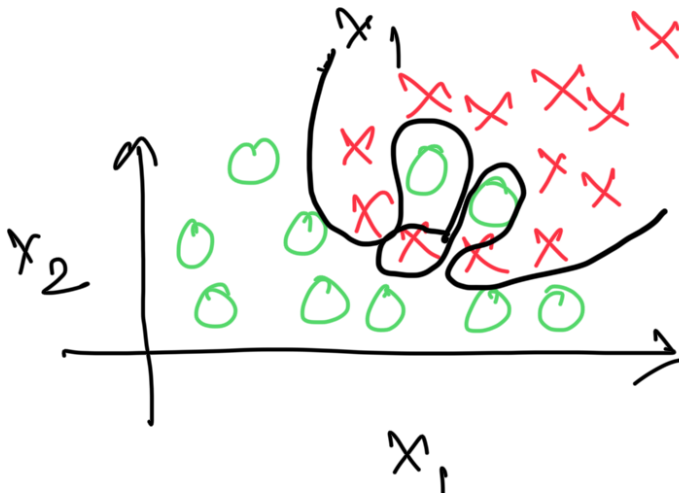
$$z = w_1 x_1 + w_2 x_2 + b$$

Underfitting -



Generalization

$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_2^2 + w_5 x_2 x_1 + b$$



Overfitting

$$z = w_1 x_1 + w_2 x_2 + w_3 x_1^2 + w_4 x_1^2 x_2 + w_5 x_1^2 x_2^3 + \dots + b$$